Figures

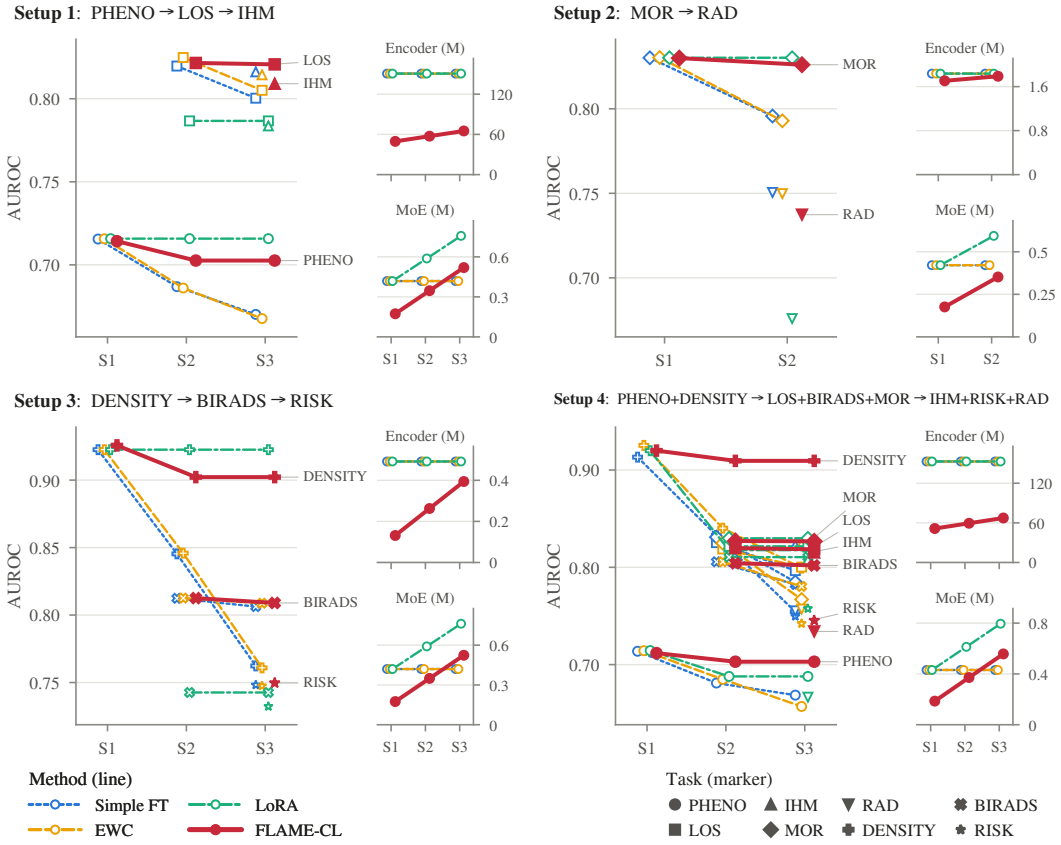


Figure 1: (Fig. 5 redraw, compact main-text version, about the height of the original.) Per-stage continual-learning trajectories for SIMPLE FT, EWC, LoRA (MoE adapters) and FLAME-CL on the four task sequences (seed 42): AUROC of every task seen so far after each stage Sk (marker = task; colour and line style = method; FLAME-CL solid), with encoder and MoE parameters at inference. For FLAME-CL the value at a task's own stage is before rank-32 truncation. Numbers in Tab. 1; AUPRC in Fig. 10.

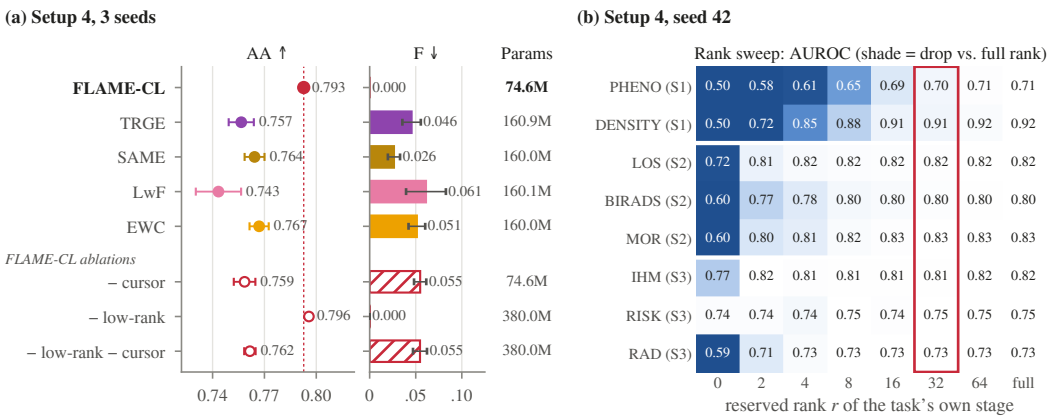


Figure 2: (Replaces Tabs. 3–4 on page 9.) (a) Setup 4, 3 timestamp seeds: average accuracy (AA, dot ± std), forgetting F (bar ± std) and parameters after three stages, for FLAME-CL, the baselines and the component ablations (hatched/hollow). (b) CL-phase rank sweep (Setup 4, seed 42): AUROC when the active component of a task's own stage is truncated to rank r, earlier stages frozen at rank 32; shade = drop relative to full rank; $r = 32$ (default) outlined. Full numbers in Tabs. 14 and 16.

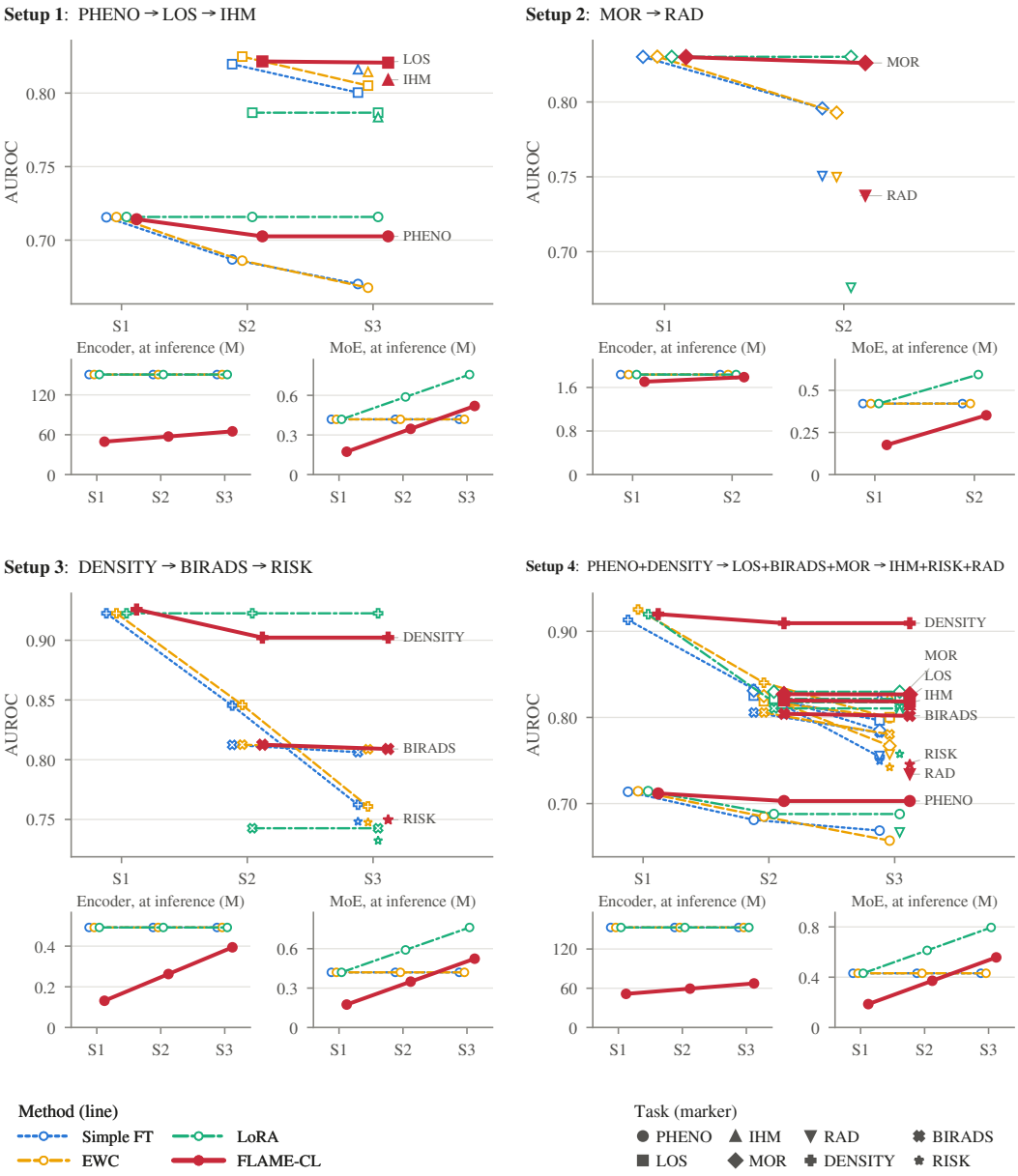


Figure 3: (Fig. 5 redraw, full-size version for an appendix or if space allows.) Same content as the compact version, with the parameter panels below each AUROC panel.

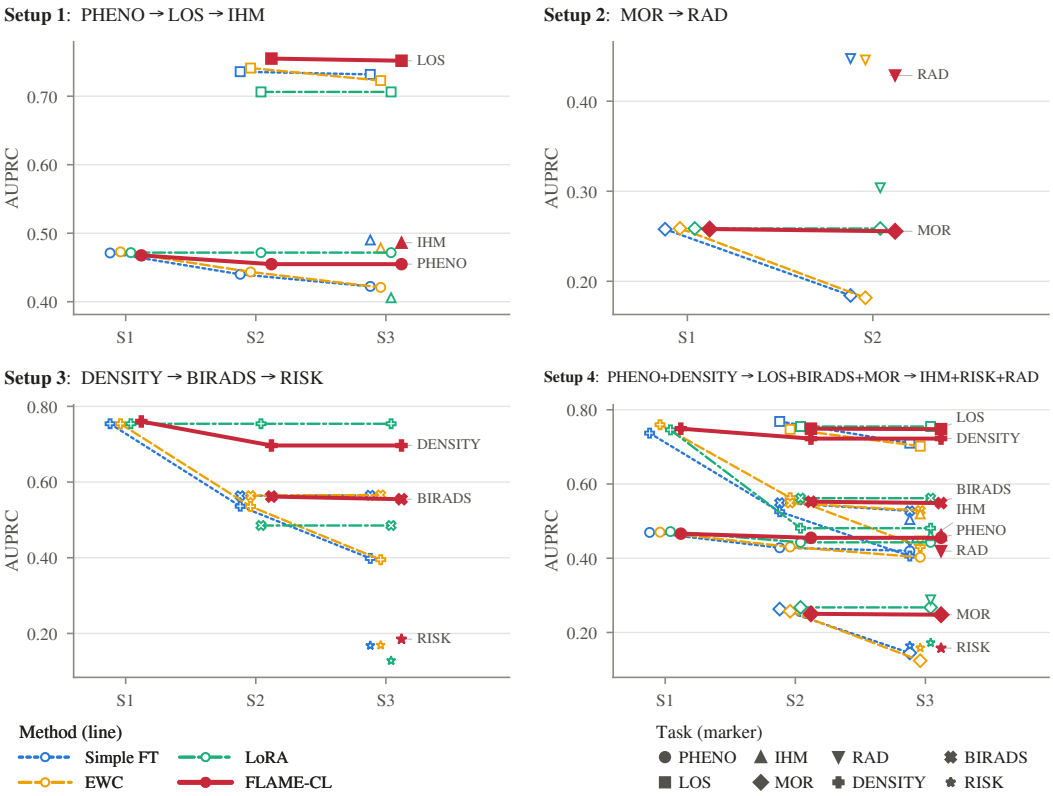


Figure 4: (Fig. 10 redraw) Per-stage AUPRC; methods and encoding as in Fig. 3.

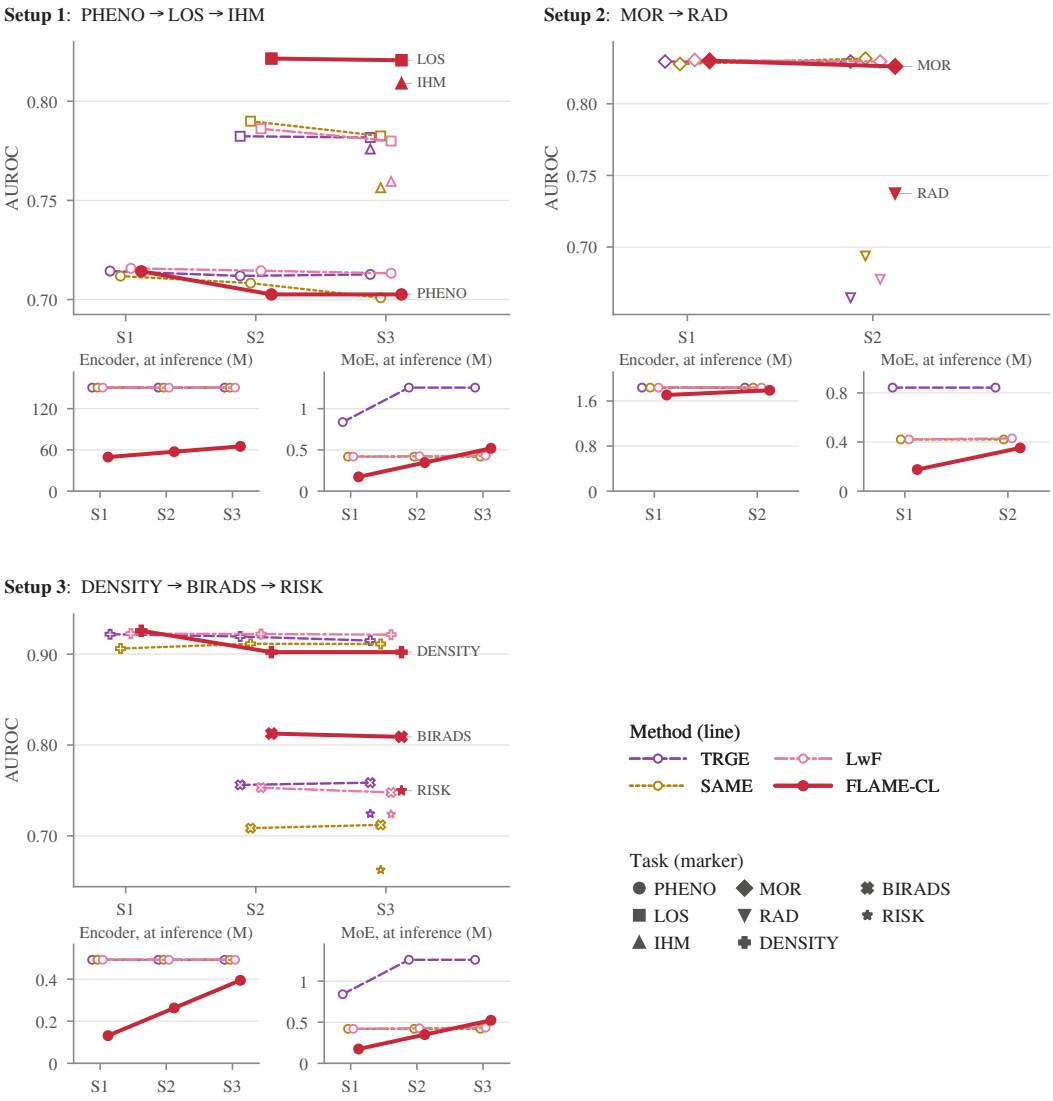


Figure 5: (New, appendix) FLAME-CL against TRGE, SAME and LwF on Setups 1–3 (seed 42). Parameter panels show inference-time totals: the three baselines adapt only the MoE and keep the full encoder. Numbers in Tab. 2.

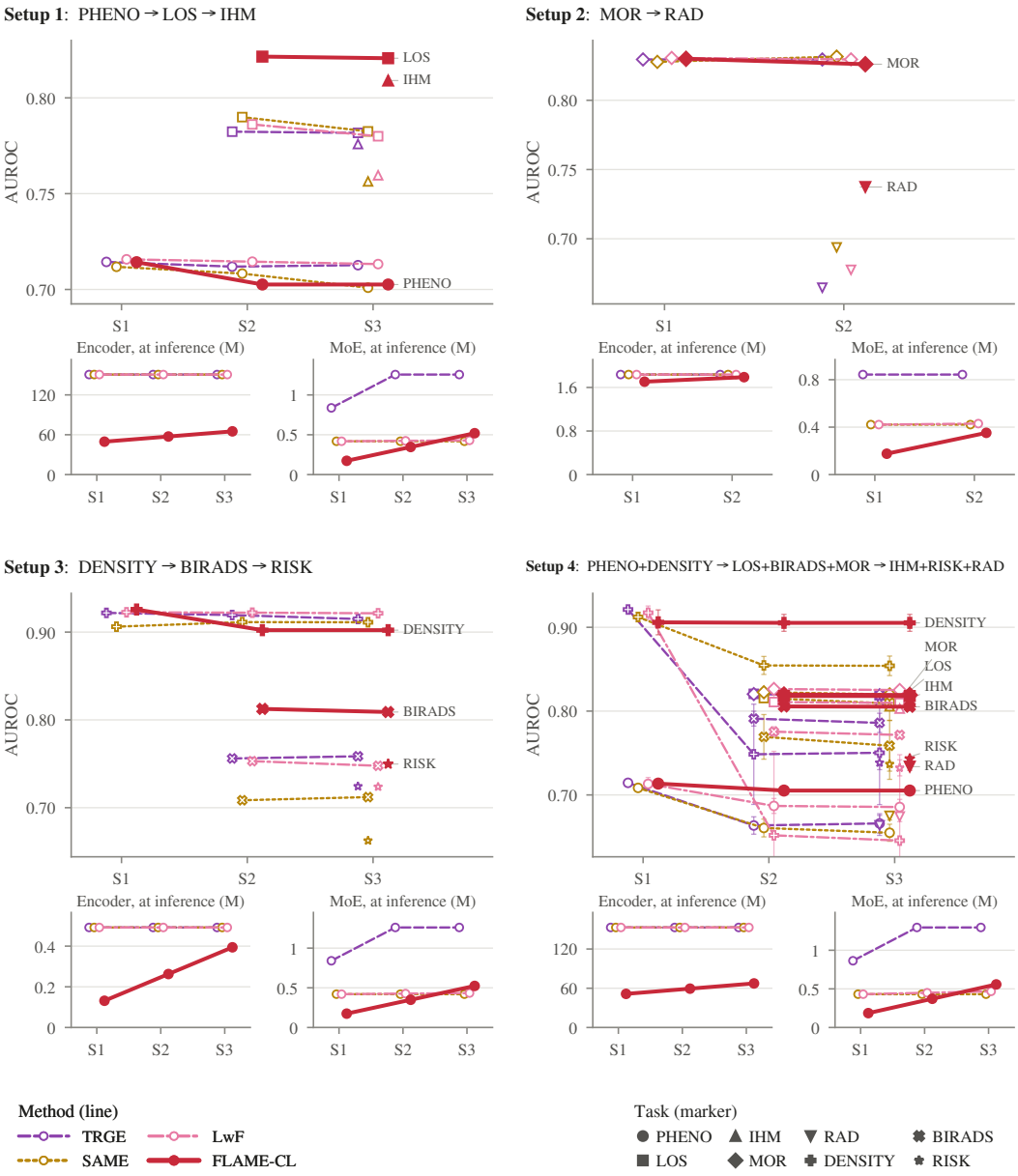


Figure 6: (Alternative) Same as above with Setup 4 added (mean ± std over the three timestamp seeds of Tabs. 3, 13, 14).

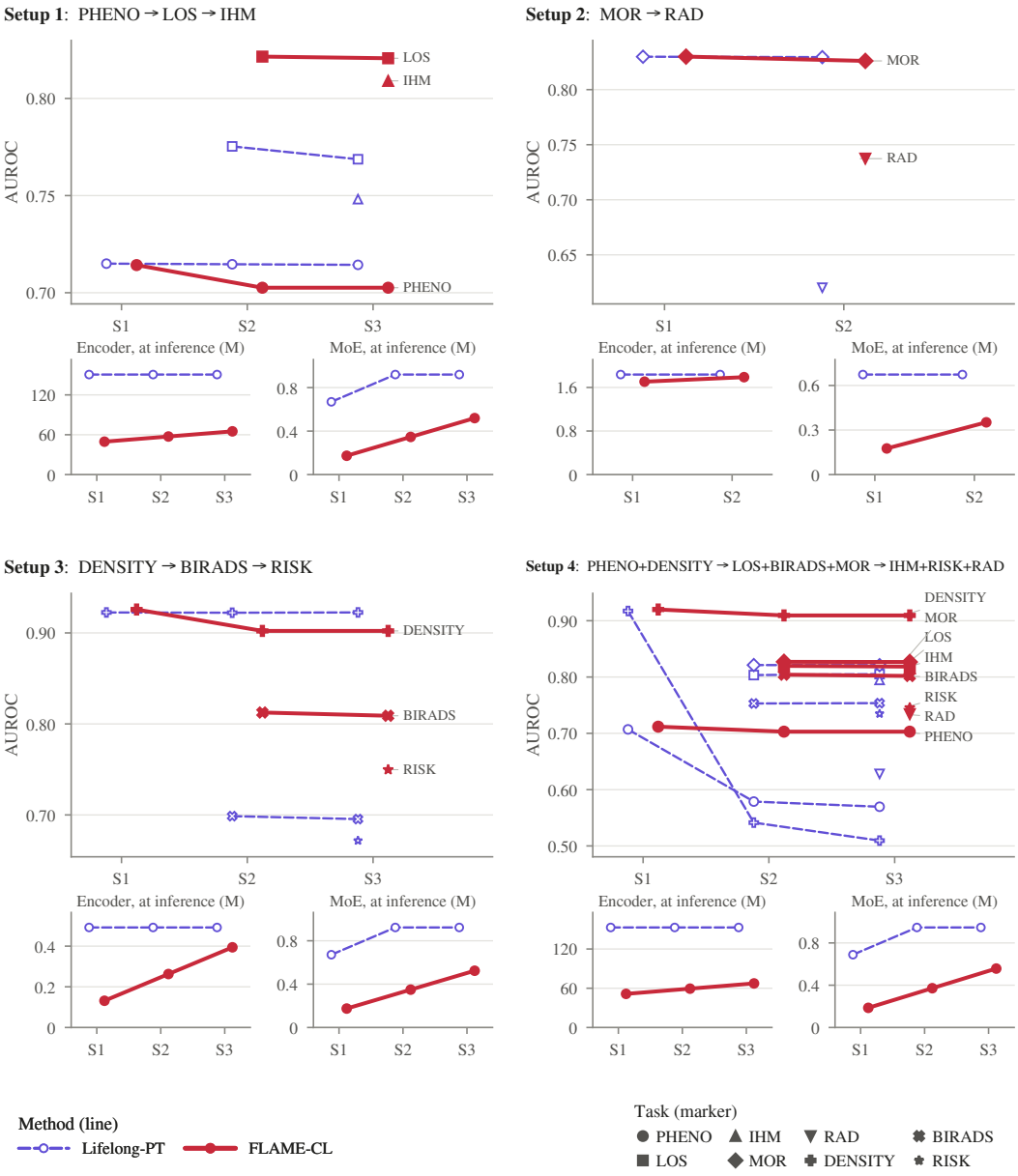


Figure 7: (Fig. 8 redraw) FLAME-CL vs. Lifelong-PT, same runs as Fig. 3; parameter panels are inference-time totals.

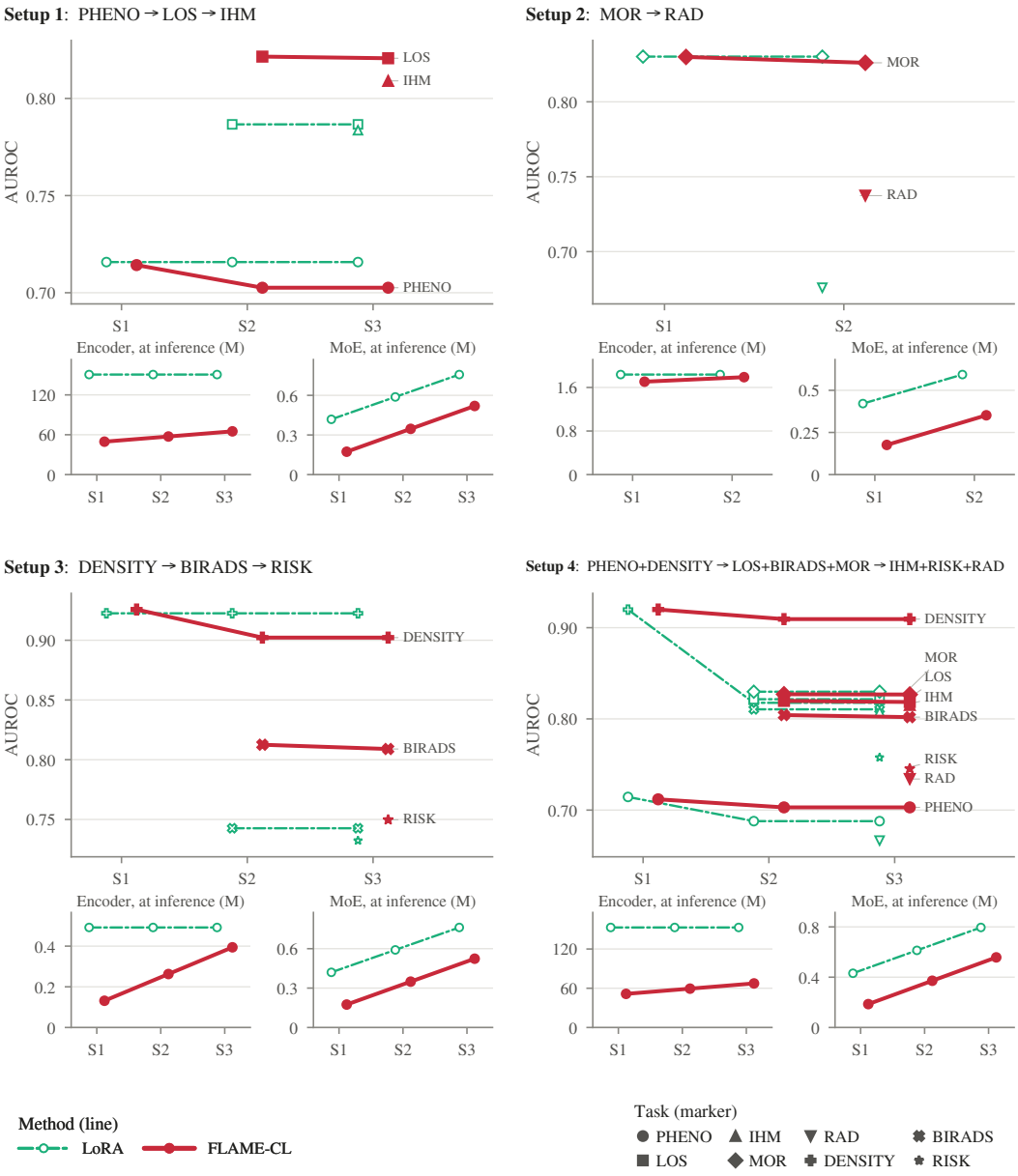


Figure 8: (Fig. 9 redraw) FLAME-CL vs. LoRA (MoE adapters); a subset of Fig. 3, so it can be dropped.

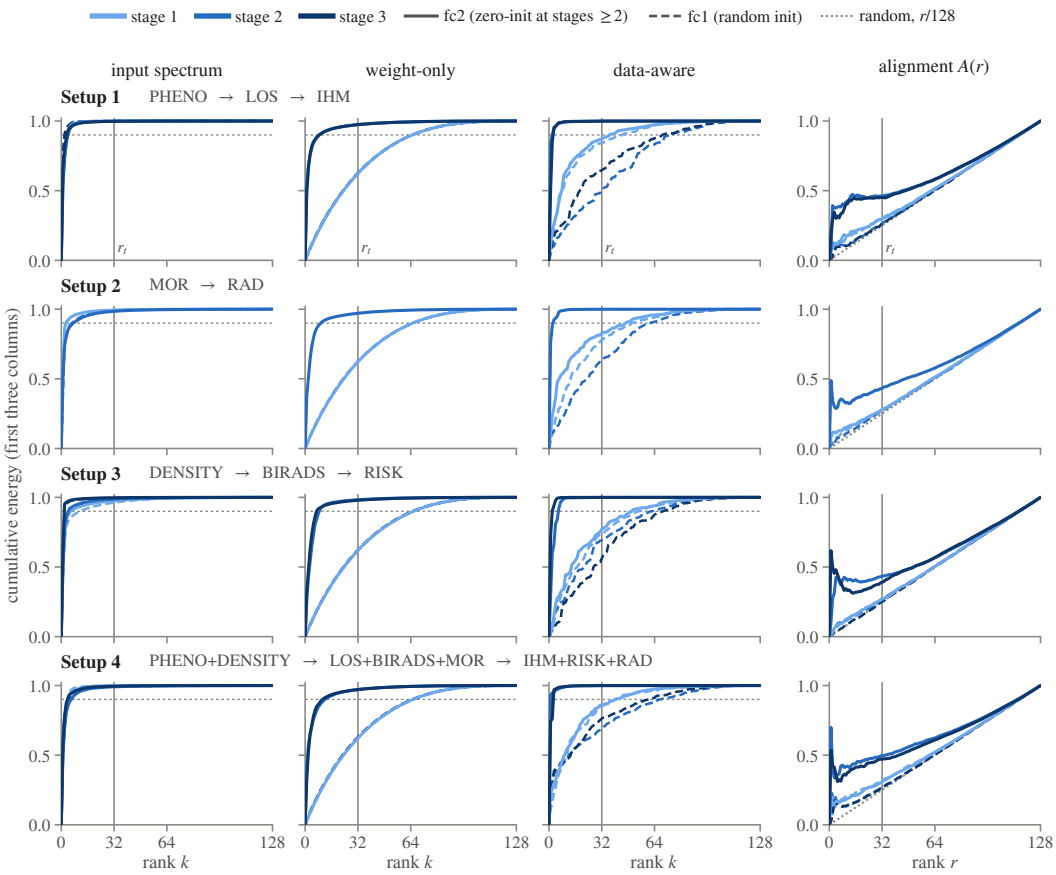


Figure 9: (New, App. D.4.3; path A, replaces Fig. 16.) Continual-stage spectra and singular-vector alignment for the four CL sequences (rows; FLAME-CL, seed 42): input spectrum, weight-only spectrum, data-aware energy, and $A(r) = \|V_r^\top Q_r\|_F^2 / r$ (dotted: random, $r/128$). Median over the 5 experts; solid fc2 (zero-initialised at stages ≥ 2), dashed fc1 (random init); colour = stage; vertical line $r_t = 32$. Numbers in Tab. 8.

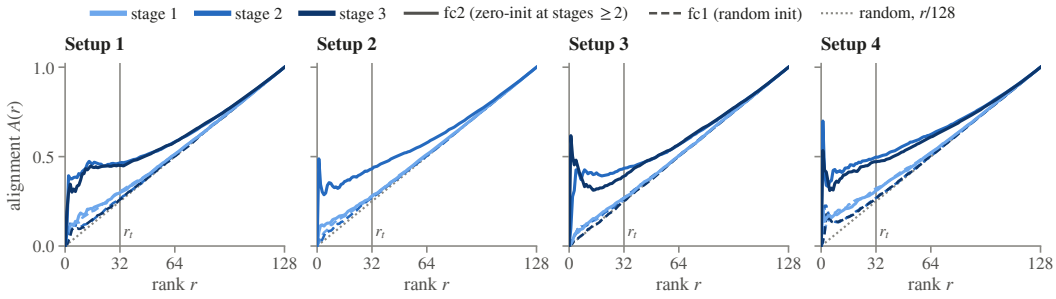


Figure 10: (New, App. D.4.3; path B, next to Fig. 16 extended with Setups 1–3.) Alignment column only.

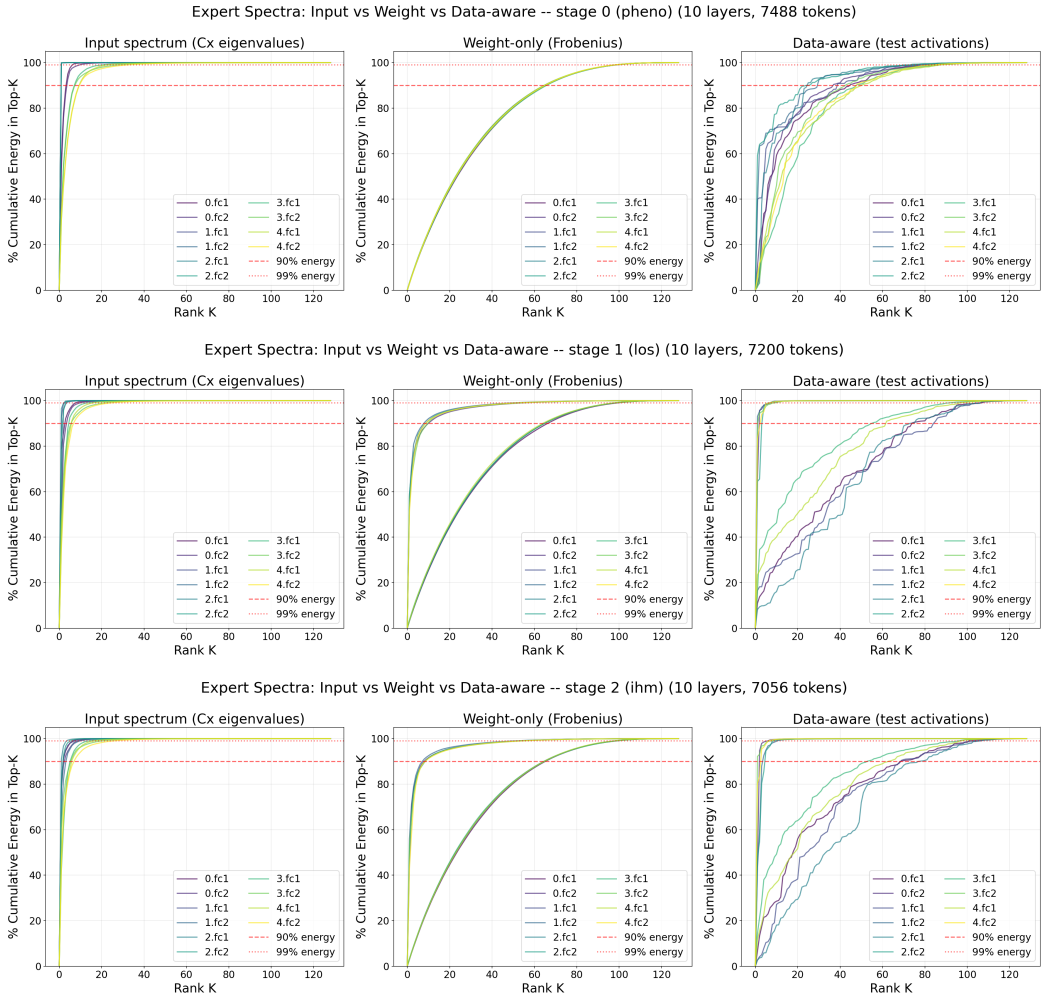


Figure 11: (Path B) Fig. 16 format for Setup 1; Setups 2–3 are `figs/fig16_setup{2,3}_stage*.png`, layout in `tables/cl_alignment_snippets.tex`.

Table 1: Per-stage numbers behind Fig. 3: AUROC when first learned → after the final stage (seed 42). Stage-K tasks report one value; drops of at least 0.02 in bold. For FLAME-CL the first value is before rank-32 truncation, so its deltas are the one-time truncation cost; F is measured against the deployed (post-truncation) model. Parameter rows give inference-time totals after the first → last stage; LoRA adapts only the MoE and keeps the full encoder.

Task (stage)	Simple FT	EWC	LoRA	FLAME-CL
Setup 1: Pheno → Los → Ihm				
PHENO (s1)	0.716 → **0.670**	0.716 → **0.668**	0.716 → 0.716	0.714 → 0.703
LOS (s2)	0.820 → 0.800	0.825 → 0.805	0.787 → 0.787	0.822 → 0.821
IHM (s3)	0.816	0.814	0.784	0.809
AA ↑	0.762	0.762	0.762	**0.777**
F ↓	0.032	0.034	0.000	0.000
Enc. params, at inference (M)	150.6	150.6	150.6	49.6→65.1
MoE params, at inference (M)	0.42	0.42	0.42→0.76	0.17→0.52
Setup 2: Mor → Rad				
MOR (s1)	0.830 → **0.796**	0.830 → **0.793**	0.830 → 0.830	0.830 → 0.826
RAD (s2)	0.751	0.750	0.676	0.737
AA ↑	0.773	0.771	0.753	**0.782**
F ↓	0.034	0.037	0.000	0.000
Enc. params, at inference (M)	1.84	1.84	1.84	1.71→1.79
MoE params, at inference (M)	0.42	0.42	0.42→0.59	0.18→0.35
Setup 3: Density → Birads → Risk				
DENSITY (s1)	0.923 → **0.762**	0.923 → **0.761**	0.923 → 0.923	0.926 → **0.902**
BIRADS (s2)	0.812 → 0.806	0.813 → 0.809	0.743 → 0.743	0.813 → 0.809
RISK (s3)	0.748	0.748	0.732	0.750
AA ↑	0.772	0.772	0.799	**0.820**
F ↓	0.083	0.083	0.000	0.000
Enc. params, at inference (M)	0.49	0.49	0.49	0.13→0.39
MoE params, at inference (M)	0.42	0.42	0.42→0.76	0.17→0.52
Setup 4: {Pheno, Density} → {Los, Birads, Mor} → {Ihm, Risk, Rad}				
PHENO (s1)	0.714 → **0.669**	0.714 → **0.657**	0.714 → **0.688**	0.712 → 0.703
DENSITY (s1)	0.913 → **0.754**	0.925 → **0.799**	0.920 → **0.818**	0.920 → 0.909
LOS (s2)	0.825 → **0.797**	0.819 → 0.800	0.822 → 0.822	0.820 → 0.819
BIRADS (s2)	0.806 → **0.782**	0.806 → **0.780**	0.810 → 0.810	0.804 → 0.802
MOR (s2)	0.831 → **0.785**	0.825 → **0.767**	0.830 → 0.830	0.827 → 0.827
IHM (s3)	0.823	0.824	0.808	0.815
RISK (s3)	0.750	0.742	0.758	0.745
RAD (s3)	0.755	0.757	0.667	0.734
AA ↑	0.764	0.766	0.775	**0.794**
F ↓	0.060	0.057	0.026	0.000
Enc. params, at inference (M)	153.0	153.0	153.0	51.4→67.4
MoE params, at inference (M)	0.43	0.43	0.43→0.80	0.19→0.56

Tables

D.6 drop-in text. Routing does react to corruption, with the Jensen–Shannon divergence from the clean gate rising with the noise level on every task, but that reaction is uncorrelated with the modality's importance (Spearman $\rho = +0.14$, $p = 0.44$, $n = 31$ (task, modality) pairs). Reliability weighting instead resides in the fusion head: the learned share of each modality in the fusion logit correlates with its measured importance (the AUROC drop on removal) at $\rho = +0.73$ ($p = 2.7\times10^{-6}$, $n = 31$), and per task at $\rho = +1$ on 5 of 8 tasks (Tab. 7).

Table 2: Setups 1–3: AUROC when first learned → after the final stage for FLAME-CL and the MoE-based / distillation baselines (seed 42; Setup 4 is in Tabs. 3, 4). Drops of at least 0.02 in bold; F against the model deployed at the stage where each task was learned. Parameter rows give inference-time totals after the first → last stage: TRGE, SAME and LwF adapt only the MoE and keep the full encoder.

Task (stage)	FLAME-CL	TRGE	SAME	LwF	EWC
Setup 1: PHENO → LOS → IHM					
PHENO (s1)	0.714 → 0.703	0.714 → 0.713	0.712 → 0.701	0.716 → 0.713	0.716 → **0.668**
LOS (s2)	0.822 → 0.821	0.782 → 0.782	0.790 → 0.782	0.786 → 0.780	0.825 → 0.805
IHM (s3)	0.809	0.776	0.756	0.760	0.814
AA ↑	**0.777**	0.757	0.747	0.751	0.762
F ↓	0.000	0.001	0.009	0.004	0.034
Enc. params, at inference (M)	49.6→65.1	150.6	150.6	150.6	150.6
MoE params, at inference (M)	0.17→0.52	0.84→1.26	0.42	0.42→0.43	0.42
Setup 2: MOR → RAD					
MOR (s1)	0.830 → 0.826	0.830 → 0.829	0.828 → 0.832	0.831 → 0.830	0.830 → **0.793**
RAD (s2)	0.737	0.665	0.694	0.678	0.750
AA ↑	**0.782**	0.747	0.763	0.754	0.771
F ↓	0.000	0.000	0.000	0.001	0.037
Enc. params, at inference (M)	1.71→1.79	1.84	1.84	1.84	1.84
MoE params, at inference (M)	0.18→0.35	0.84	0.42	0.42→0.43	0.42
Setup 3: DENSITY → BIRADS → RISK					
DENSITY (s1)	0.926 → **0.902**	0.922 → 0.915	0.906 → 0.911	0.923 → 0.921	0.923 → **0.761**
BIRADS (s2)	0.813 → 0.809	0.756 → 0.759	0.708 → 0.712	0.753 → 0.748	0.813 → 0.809
RISK (s3)	0.750	0.724	0.663	0.724	0.748
AA ↑	**0.820**	0.799	0.762	0.798	0.772
F ↓	0.000	0.003	0.000	0.003	0.083
Enc. params, at inference (M)	0.13→0.39	0.49	0.49	0.49	0.49
MoE params, at inference (M)	0.17→0.52	0.84→1.26	0.42	0.42→0.43	0.42

Table 3: Setup 4 per-task AUROC when first learned → after the final stage (3 timestamp-generated seeds, mean ± std). Stage-3 tasks are learned last and report one value. Drops of at least 0.02 in bold.

Task (stage)	FLAME-CL	TRGE	SAME	LwF	EWC
PHENO (s1)	$0.71_{\pm 0.00} \rightarrow 0.71_{\pm 0.00}$	$0.71_{\pm 0.00} \rightarrow \mathbf{0.67}_{\pm 0.01}$	$0.71_{\pm 0.00} \rightarrow \mathbf{0.65}_{\pm 0.01}$	$0.71_{\pm 0.01} \rightarrow \mathbf{0.69}_{\pm 0.01}$	$0.71_{\pm 0.00} \rightarrow \mathbf{0.67}_{\pm 0.00}$
DENSITY (s1)	$0.91_{\pm 0.01} \rightarrow 0.91_{\pm 0.01}$	$0.92_{\pm 0.00} \rightarrow \mathbf{0.75}_{\pm 0.06}$	$0.91_{\pm 0.01} \rightarrow \mathbf{0.85}_{\pm 0.01}$	$0.92_{\pm 0.01} \rightarrow \mathbf{0.65}_{\pm 0.10}$	$0.92_{\pm 0.01} \rightarrow \mathbf{0.78}_{\pm 0.05}$
LOS (s2)	$0.82_{\pm 0.00} \rightarrow 0.82_{\pm 0.00}$	$0.82_{\pm 0.00} \rightarrow 0.82_{\pm 0.00}$	$0.82_{\pm 0.00} \rightarrow 0.81_{\pm 0.01}$	$0.81_{\pm 0.01} \rightarrow 0.81_{\pm 0.00}$	$0.82_{\pm 0.00} \rightarrow 0.80_{\pm 0.01}$
BIRADS (s2)	$0.81_{\pm 0.00} \rightarrow 0.81_{\pm 0.00}$	$0.79_{\pm 0.01} \rightarrow 0.79_{\pm 0.01}$	$0.77_{\pm 0.03} \rightarrow 0.76_{\pm 0.03}$	$0.78_{\pm 0.01} \rightarrow 0.77_{\pm 0.00}$	$0.80_{\pm 0.00} \rightarrow 0.79_{\pm 0.01}$
MOR (s2)	$0.82_{\pm 0.00} \rightarrow 0.82_{\pm 0.00}$	$0.82_{\pm 0.00} \rightarrow 0.82_{\pm 0.00}$	$0.82_{\pm 0.00} \rightarrow 0.82_{\pm 0.00}$	$0.83_{\pm 0.00} \rightarrow 0.83_{\pm 0.00}$	$0.82_{\pm 0.00} \rightarrow \mathbf{0.78}_{\pm 0.02}$
IHM (s3)	$0.81_{\pm 0.01}$	$0.81_{\pm 0.00}$	$0.81_{\pm 0.02}$	$0.80_{\pm 0.00}$	$0.81_{\pm 0.00}$
RISK (s3)	$0.74_{\pm 0.00}$	$0.74_{\pm 0.01}$	$0.74_{\pm 0.02}$	$0.73_{\pm 0.01}$	$0.75_{\pm 0.00}$
RAD (s3)	$0.73_{\pm 0.00}$	$0.66_{\pm 0.01}$	$0.68_{\pm 0.00}$	$0.67_{\pm 0.01}$	$0.75_{\pm 0.00}$

Table 4: Setup 4 per-task forgetting F and summary metrics (3 timestamp-generated seeds, mean ± std). F is measured against the model deployed at the stage where each task was learned (post-truncation for FLAME-CL).

	FLAME-CL	TRGE	SAME	LwF	EWC
F: PHENO (s1)	0.000 ± 0.000	0.048 ± 0.009	0.054 ± 0.007	0.028 ± 0.002	0.043 ± 0.005
F: DENSITY (s1)	0.000 ± 0.000	0.171 ± 0.061	0.058 ± 0.011	0.272 ± 0.109	0.143 ± 0.054
F: LOS (s2)	0.000 ± 0.000	0.002 ± 0.000	0.006 ± 0.005	0.001 ± 0.001	0.020 ± 0.011
F: BIRADS (s2)	0.000 ± 0.000	0.005 ± 0.004	0.011 ± 0.010	0.004 ± 0.002	0.009 ± 0.011
F: MOR (s2)	0.000 ± 0.000	0.001 ± 0.001	0.003 ± 0.002	0.001 ± 0.000	0.041 ± 0.021
F (avg) ↓	0.000 ± 0.000	0.046 ± 0.010	0.026 ± 0.007	0.061 ± 0.022	0.051 ± 0.009
BWT (avg) ↑	0.000 ± 0.000	−0.046 ± 0.010	−0.026 ± 0.007	−0.061 ± 0.022	−0.051 ± 0.009
AA (8 tasks) ↑	0.793 ± 0.002	0.757 ± 0.007	0.764 ± 0.006	0.743 ± 0.013	0.767 ± 0.006

Table 5: Per-task AUROC under the router-balancing loss weight sweep (joint nine-task training, $\lambda_{\text{int}} = 0.1$; mean ± std over seeds 42, 20492, 28411). † default. Bold: best weight per task. Range: max − min of the task mean across this sweep.

		λ_{bal}					
Dataset	Task	0	0.001	0.01†	0.1	1	Range
MIMIC-IV	48-IHM	0.801±0.013	0.806±0.009	0.809±0.004	0.806±0.007	**0.815±0.005**	0.013
	LOS	0.812±0.007	0.808±0.003	**0.817±0.002**	0.812±0.005	0.811±0.007	0.009
	25-PHENO	0.715±0.003	**0.718±0.001**	0.718±0.001	0.716±0.002	0.712±0.001	0.006
eICU	MOR	0.838±0.007	0.839±0.007	0.839±0.005	0.837±0.002	**0.839±0.006**	0.003
	RAD	0.759±0.003	**0.760±0.002**	0.759±0.001	0.754±0.006	0.758±0.005	0.006
EMBED	BIRADS	**0.809±0.004**	0.806±0.001	0.804±0.003	0.804±0.003	0.803±0.003	0.006
	RISK	0.747±0.004	0.752±0.004	0.752±0.006	**0.753±0.006**	0.750±0.005	0.006
	DENSITY	0.914±0.012	0.914±0.003	0.915±0.003	0.910±0.006	**0.916±0.001**	0.006
ADNI	DIAG	0.755±0.033	**0.788±0.020**	0.771±0.044	0.765±0.026	0.751±0.029	0.036
Mean (9 tasks)		0.795±0.001	**0.799±0.002**	0.798±0.005	0.795±0.003	0.795±0.003	0.004

Table 6: Per-task AUROC under the interaction loss weight sweep (joint nine-task training, $\lambda_{\text{bal}} = 0.01$; mean ± std over seeds 42, 20492, 28411). † default. Bold: best weight per task. Range: max − min of the task mean across this sweep.

		λ_{int}				
Dataset	Task	0	0.01	0.1†	1	Range
MIMIC-IV	48-IHM	0.806±0.006	0.808±0.004	0.809±0.004	**0.816±0.006**	0.010
	LOS	0.814±0.008	0.808±0.003	0.817±0.002	**0.817±0.006**	0.009
	25-PHENO	0.713±0.002	**0.718±0.001**	0.718±0.001	0.712±0.003	0.006
eICU	MOR	0.835±0.005	0.839±0.005	0.839±0.005	**0.840±0.006**	0.005
	RAD	0.757±0.002	0.757±0.006	0.759±0.001	**0.762±0.004**	0.006
EMBED	BIRADS	0.804±0.005	0.801±0.004	0.804±0.003	**0.806±0.002**	0.006
	RISK	0.747±0.004	0.749±0.009	**0.752±0.006**	0.748±0.001	0.005
	DENSITY	**0.917±0.003**	0.910±0.009	0.915±0.003	0.915±0.007	0.007
ADNI	DIAG	0.770±0.027	0.762±0.028	**0.771±0.044**	0.762±0.031	0.009
Mean (9 tasks)		0.796±0.004	0.795±0.002	**0.798±0.005**	0.798±0.002	0.003

Table 7: Rank correlations behind App. **??** (Setup 4 FLAME-CL, 31 (task, modality) pairs, 186 corrupted evaluations). Importance is the AUROC lost when the modality is removed; routing reaction is the Jensen–Shannon divergence of the gate distribution of a modality from its clean one. Per-task p-values are exact permutation tests (n = 3–5 modalities).

Pooled over all pairs	Spearman ρ	p	Pearson r
Routing JSD at 2× noise vs. importance	+0.14	0.44	−0.14
Routing JSD at 4× noise vs. importance	+0.06	0.75	−0.19
Routing JSD under shuffling vs. importance	+0.12	0.53	+0.27
Routing JSD at 2× noise vs. contribution share	+0.13	0.48	−0.21
Fusion contribution share vs. importance	+0.73	2.7×10^{-6}	+0.80
Fusion magnitude share vs. importance	+0.53	2.4×10^{-3}	+0.79

Share vs. importance, per task	PHENO	DENSITY	LOS	BIRADS	MOR	IHM	RISK	RAD
Modalities n	3	4	3	4	5	3	4	5
Spearman ρ	+1.00	+0.80	+1.00	+1.00	+1.00	+1.00	+0.00	+0.20
Exact p	0.33	0.33	0.33	0.08	0.02	0.33	1.00	0.78
Smallest attainable p	0.33	0.08	0.33	0.08	0.02	0.33	0.08	0.02

Table 8: Continual-phase spectra and singular-vector alignment of the active component of each stage (FLAME-CL, seed 42), median over the 5 experts, reported as fc2 / fc1. r_{90}: rank at 90% cumulative energy (ambient 128). Kept: data-aware energy inside the top-32 singular directions, i.e. what survives the r_t=32 truncation. $A(r) = \|V_r^\top Q_r\|_F^2 / r$ between the top-r right singular vectors of the component and the top-r eigenvectors of its routed-input second moment; random subspaces give $r/128$ (0.125 at r=16, 0.25 at r=32). fc2 is zero-initialised at stages ≥2; fc1, and both sublayers at stage 1, start from random init.

		r_{90} (fc2 / fc1)			Kept at r_t=32	$A(16)$
Setup	Stage	input	weight-only	data-aware	data-aware	(random 0.125)
1	1: PHENO	4 / 4	65 / 66	37 / 45	0.87 / 0.84	0.21 / 0.20
	2: LOS	5 / 3	9 / 65	2 / 74	1.00 / 0.51	0.46 / 0.14
	3: IHM	4 / 2	9 / 65	3 / 69	1.00 / 0.65	0.44 / 0.15
2	1: MOR	3 / 7	66 / 66	47 / 50	0.83 / 0.78	0.17 / 0.17
	2: RAD	8 / 7	10 / 66	3 / 62	1.00 / 0.64	0.36 / 0.15
3	1: DENSITY	6 / 12	66 / 65	50 / 55	0.77 / 0.74	0.16 / 0.15
	2: BIRADS	5 / 6	9 / 66	6 / 62	1.00 / 0.69	0.41 / 0.16
	3: RISK	2 / 2	7 / 66	2 / 69	1.00 / 0.56	0.32 / 0.12
4	1: PHENO+DENSITY	4 / 4	66 / 66	41 / 41	0.86 / 0.85	0.21 / 0.23
	2: LOS+BIRADS+MOR	7 / 5	12 / 66	1 / 68	1.00 / 0.69	0.44 / 0.16
	3: IHM+RISK+RAD	4 / 5	11 / 65	3 / 61	1.00 / 0.76	0.41 / 0.16